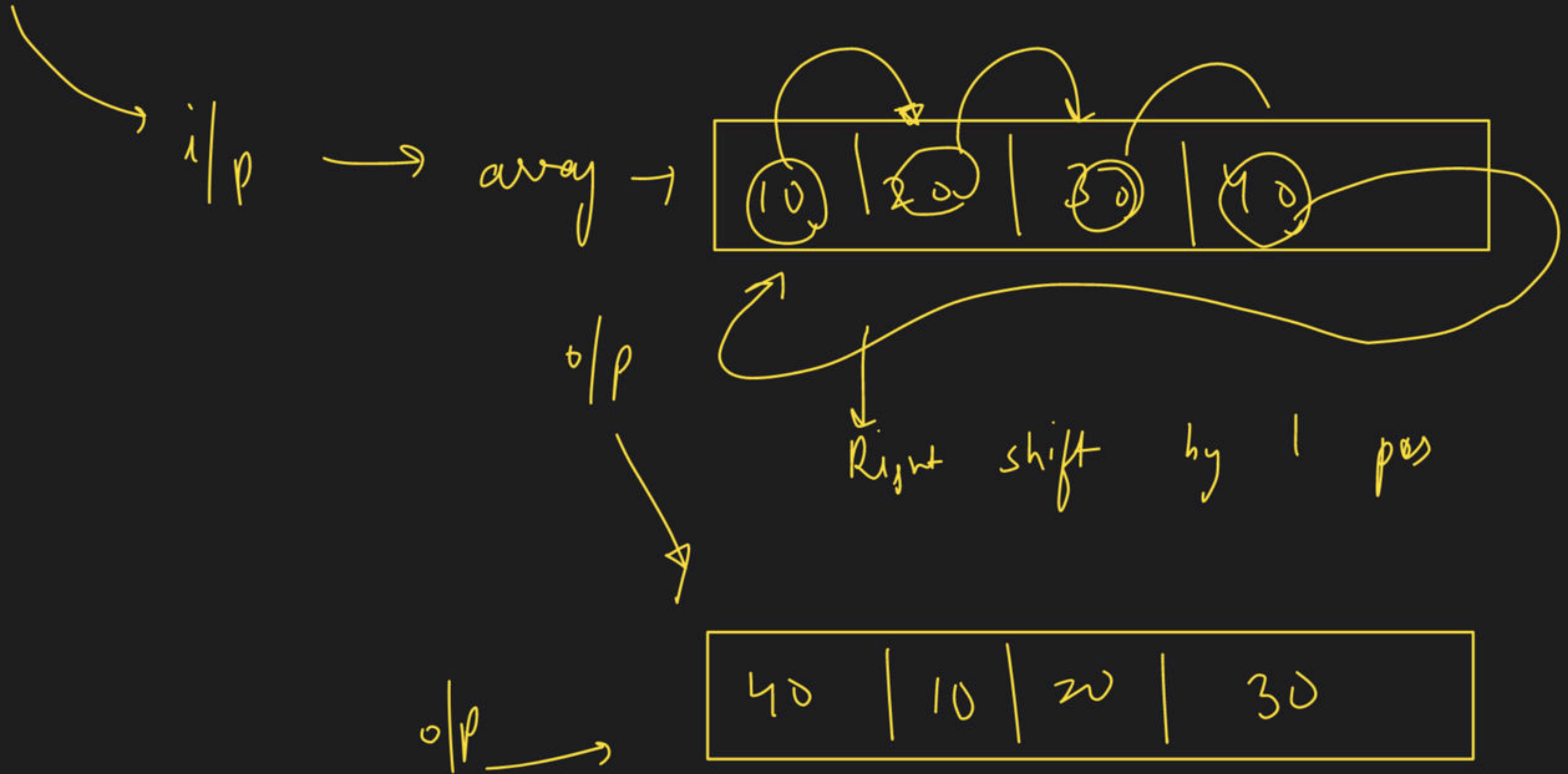


L3-Arrays

Special class

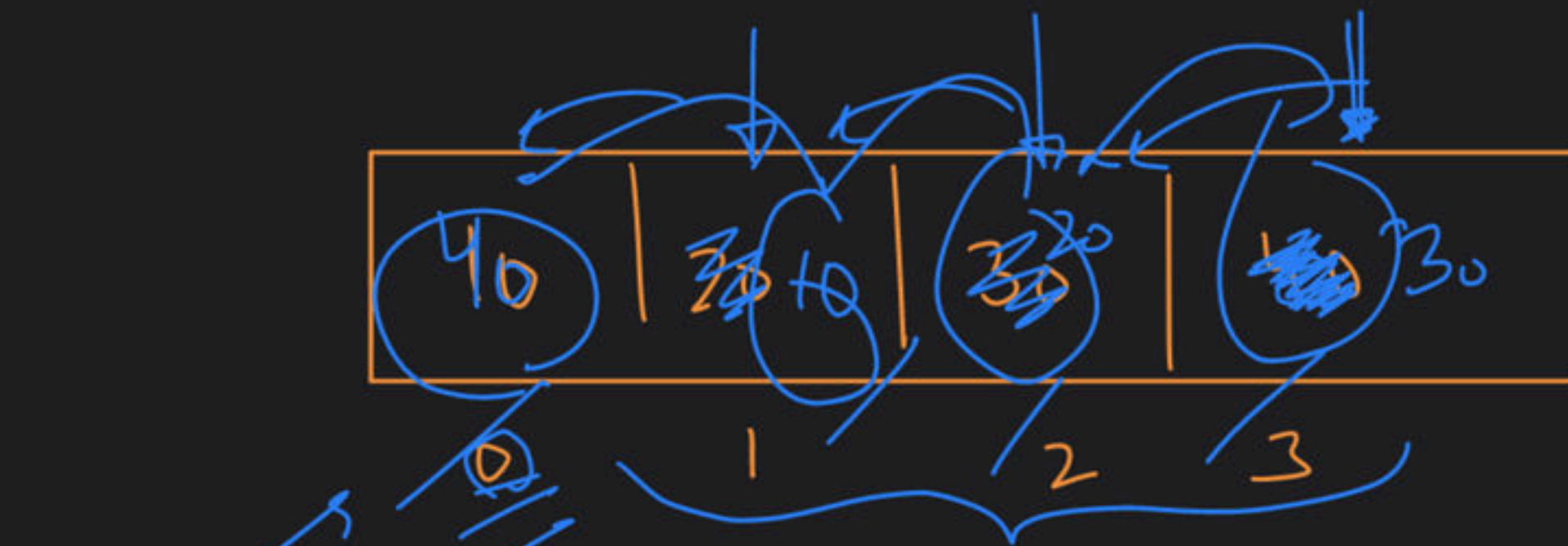
cpp stl
↳ Algorithms ✓✓

Shift the elements of Array to Right by 1 position



Approach :-

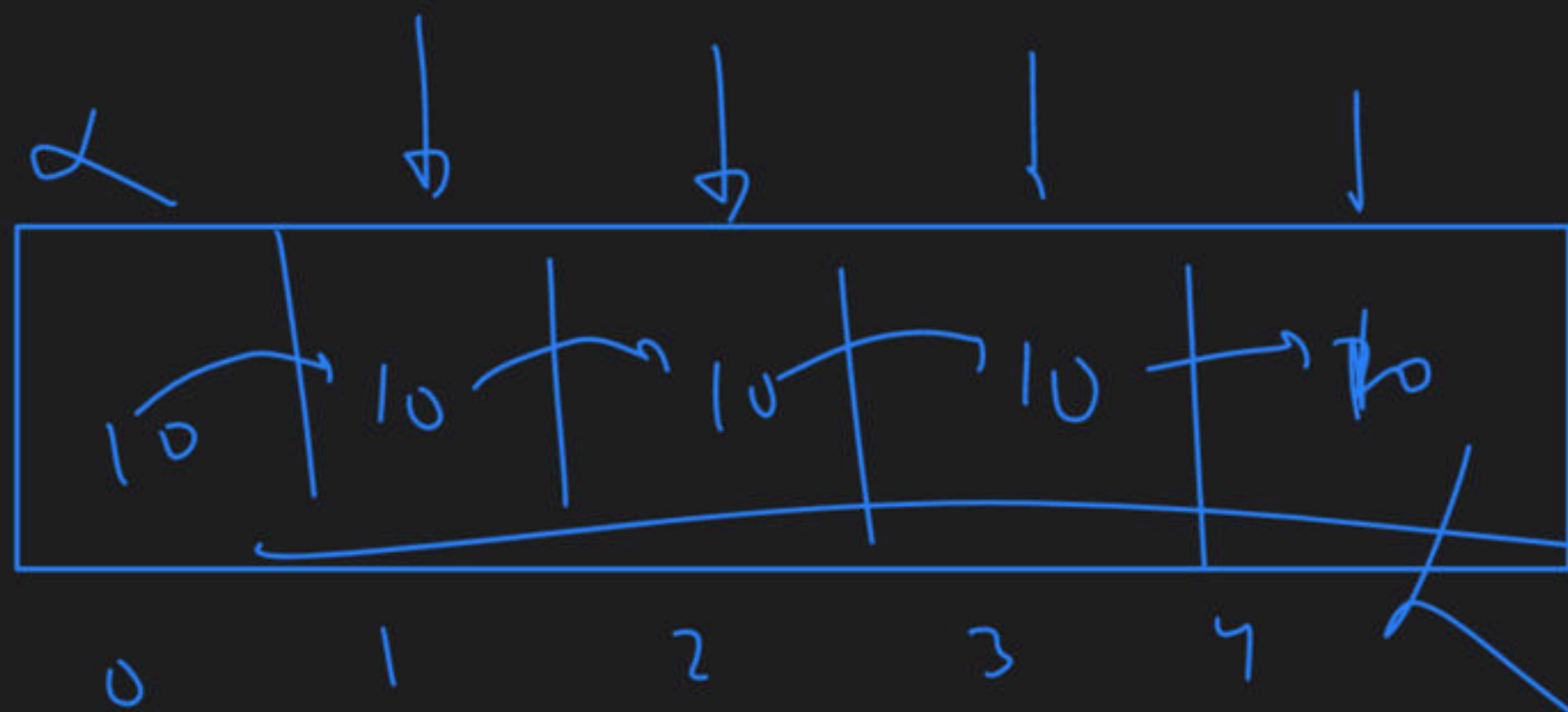
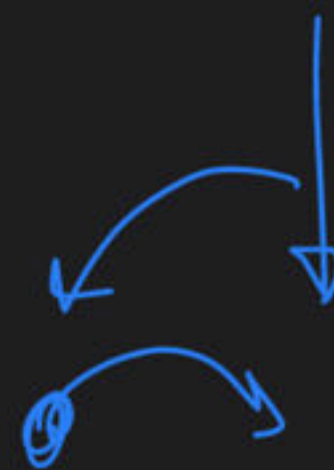
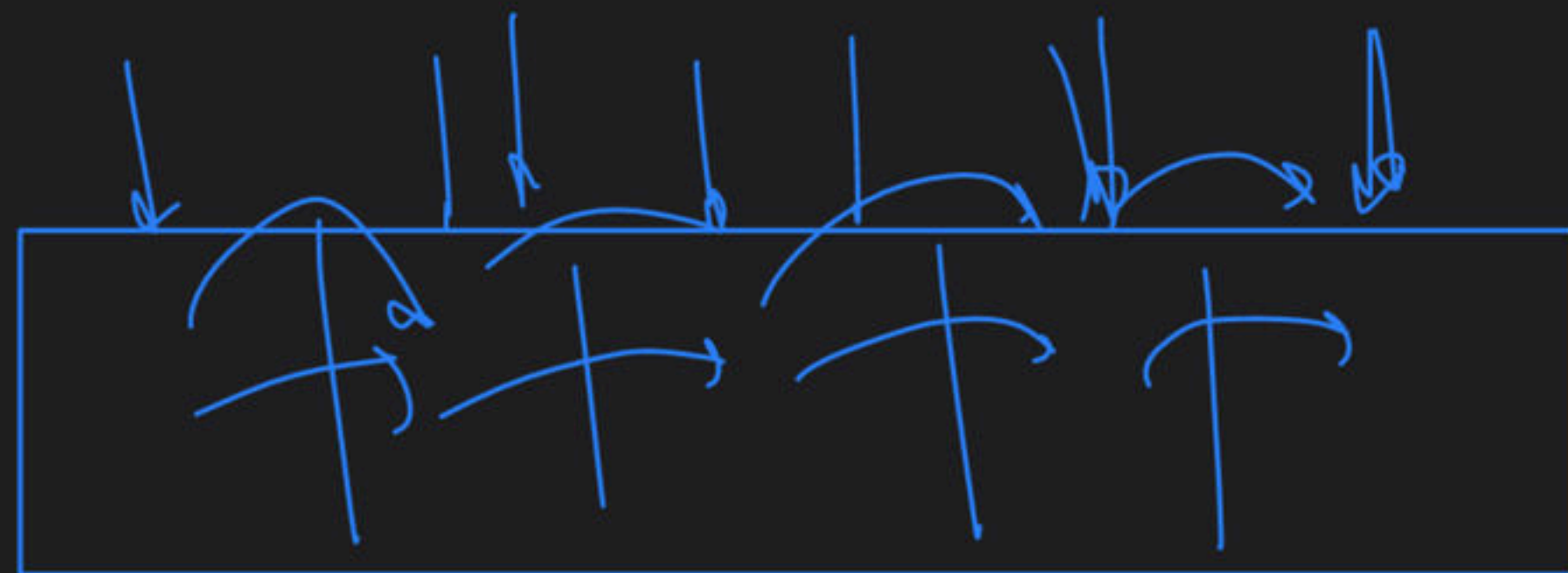
```
last = arr[n-1]
for (n-1 → 0)
{
    if (i != 0)
        arr[i] = arr[i-1]
    else
        arr[0] = last;
}
```



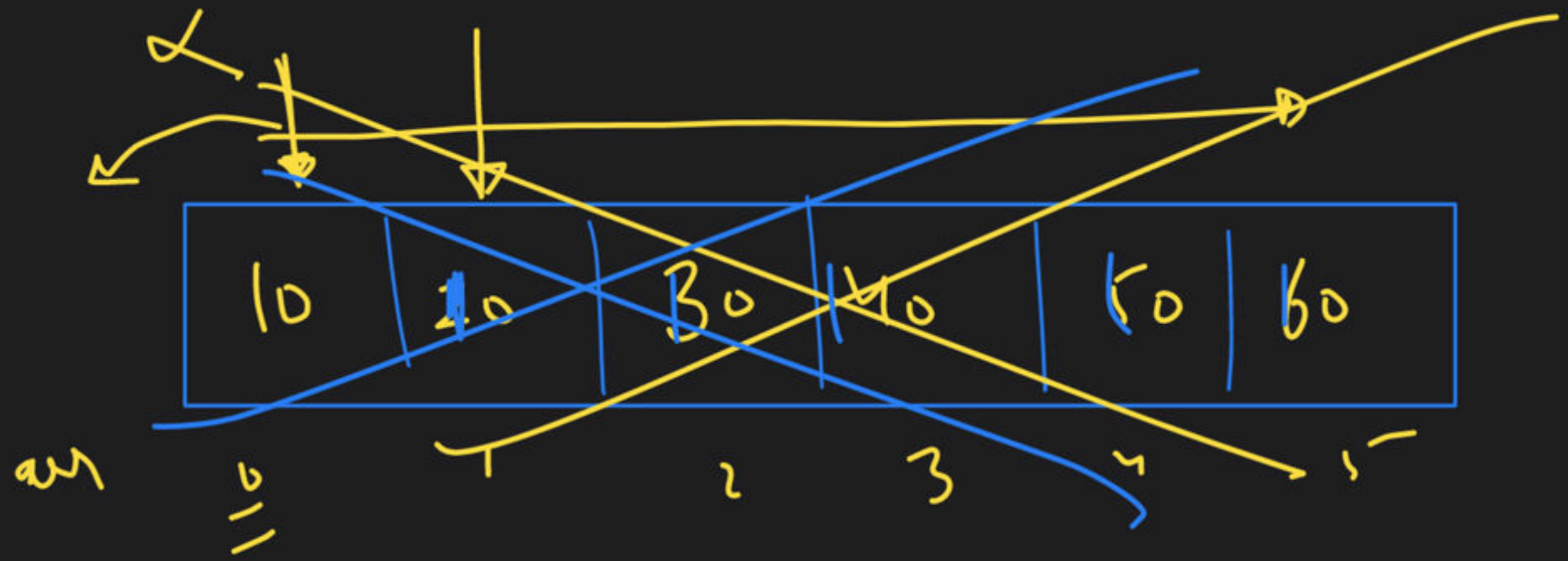
③ → 0
n-1 → 0



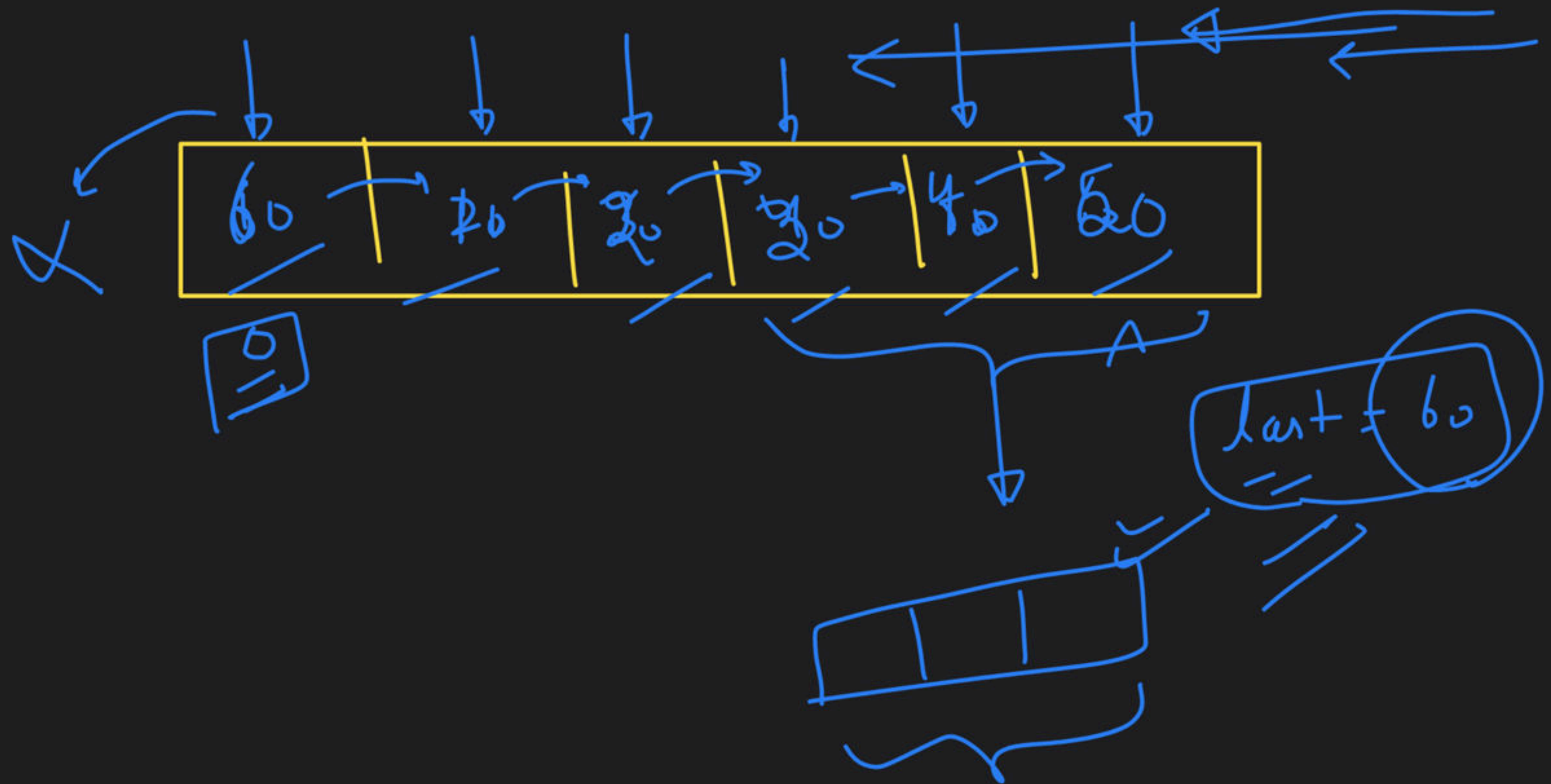
temp location / storage variable



for (0 to n)



081



1 element $\rightarrow$ $<$

n element shift

Rotate an Array $\rightarrow$ $K=3$

hilo

10	20	30	40	50	60	70
----	----	----	----	----	----	----

arr

0

1

2

3

4

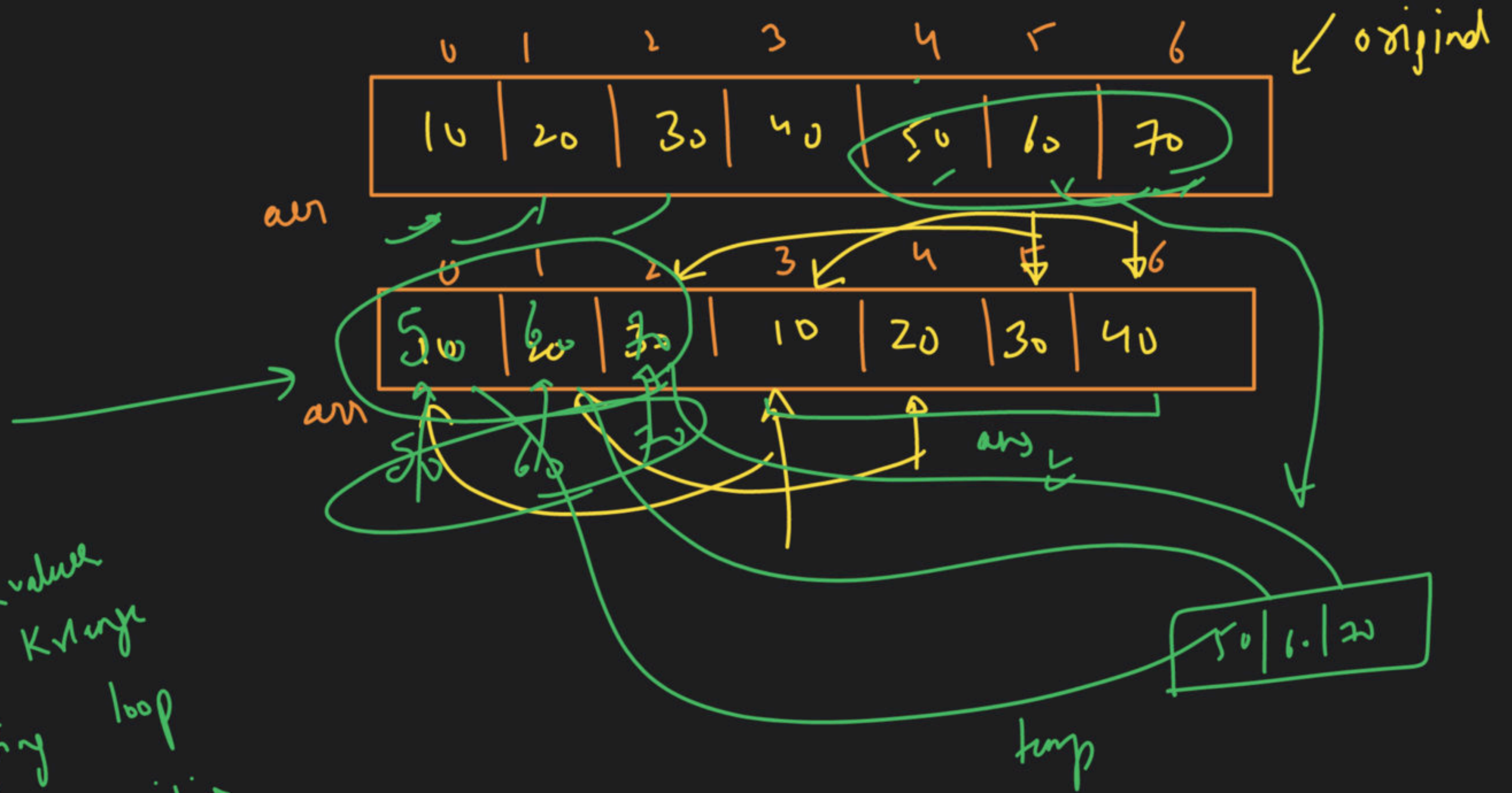
5

6

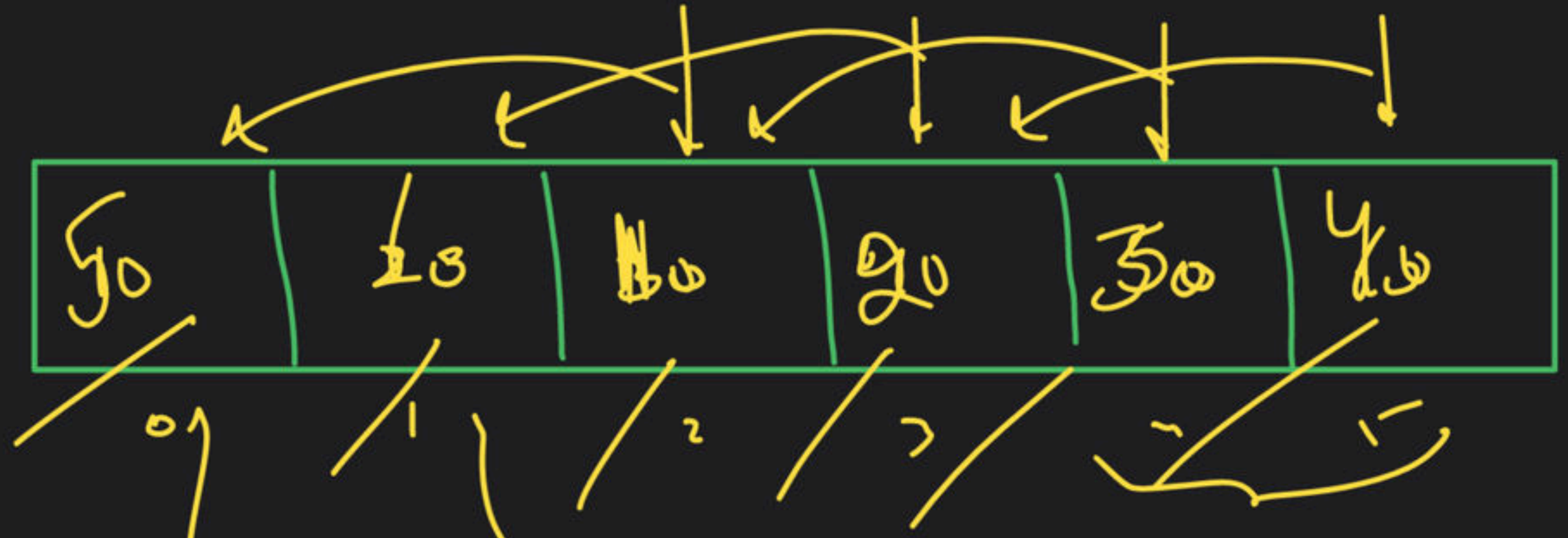
50 60 70

Approach

- (1) last K values
store K range
- (2) shifting loop
- (3) start K position
→ heap as copy



K=2



n-1

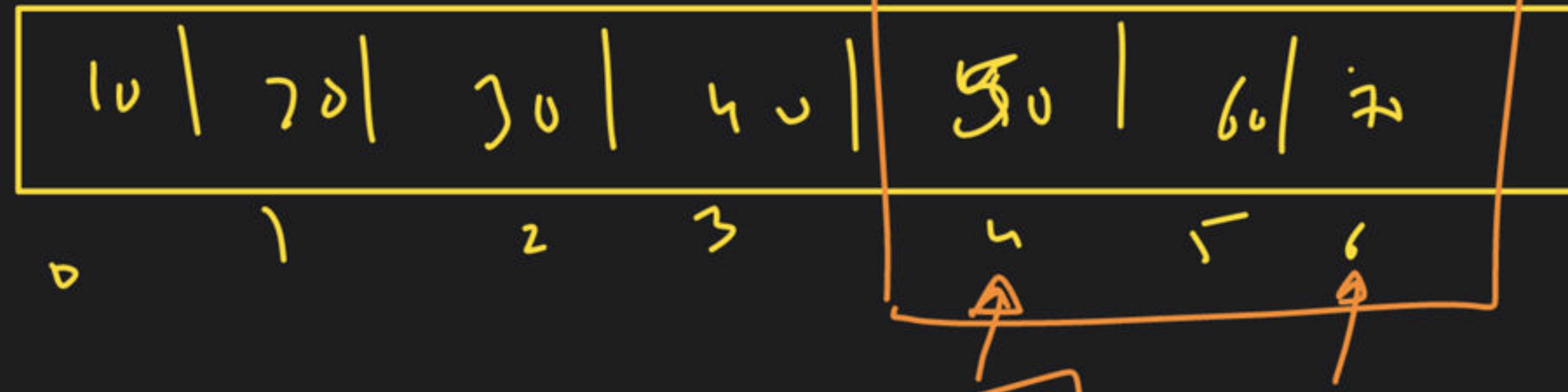
n-1

O(n-1)

O(n)

50 60

temp



$$k=3$$

$$n=7$$

$O(1)$ $\rightarrow$ space $\downarrow$
Now

$$n-k$$

$$n-1$$

$$n-k \rightarrow (n-1)$$



Cpp STL

Container

Vector

vector

Shubham



vector

Ramul

Surul

MyKish

Rut

2-D Array

n

m

row/col

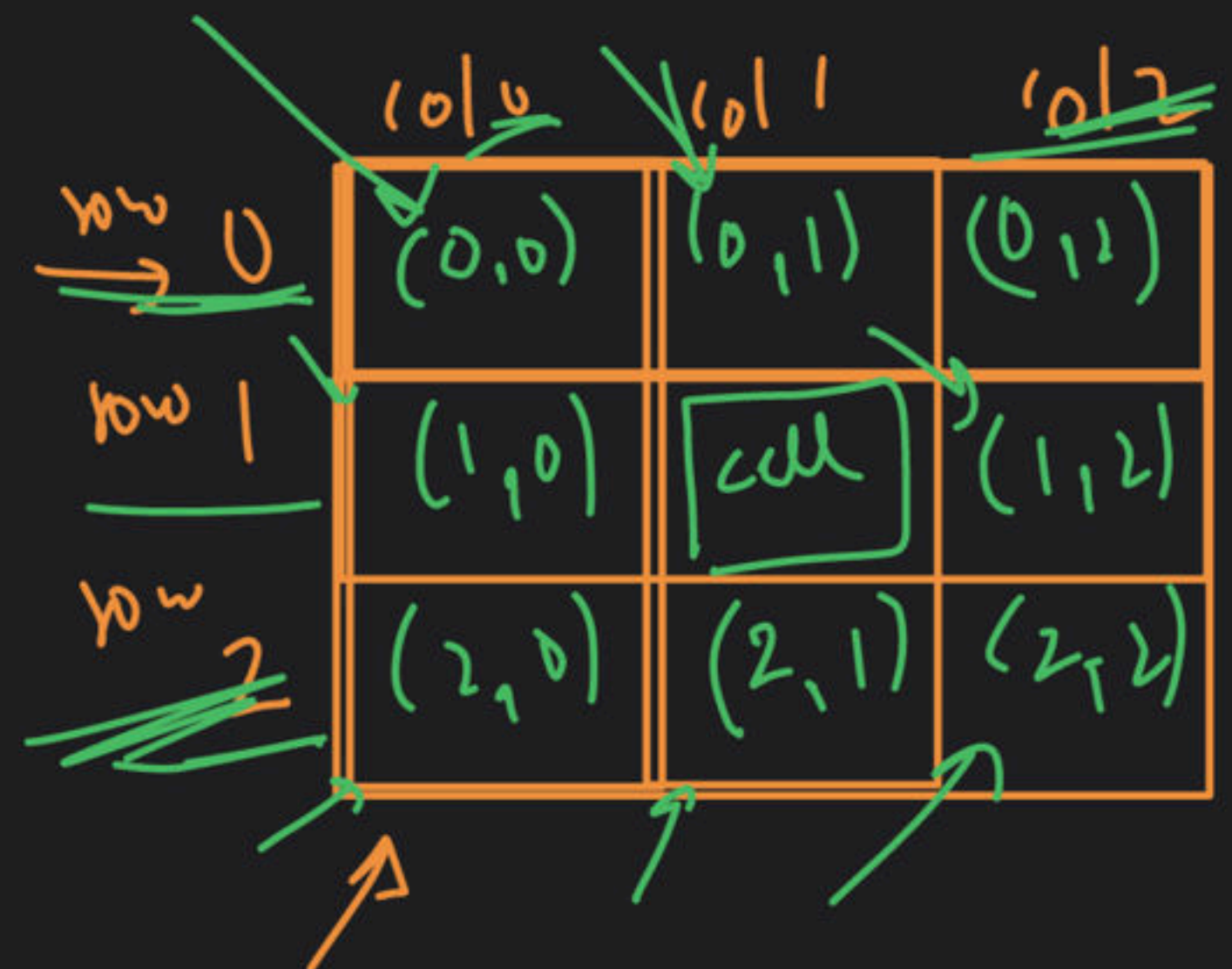
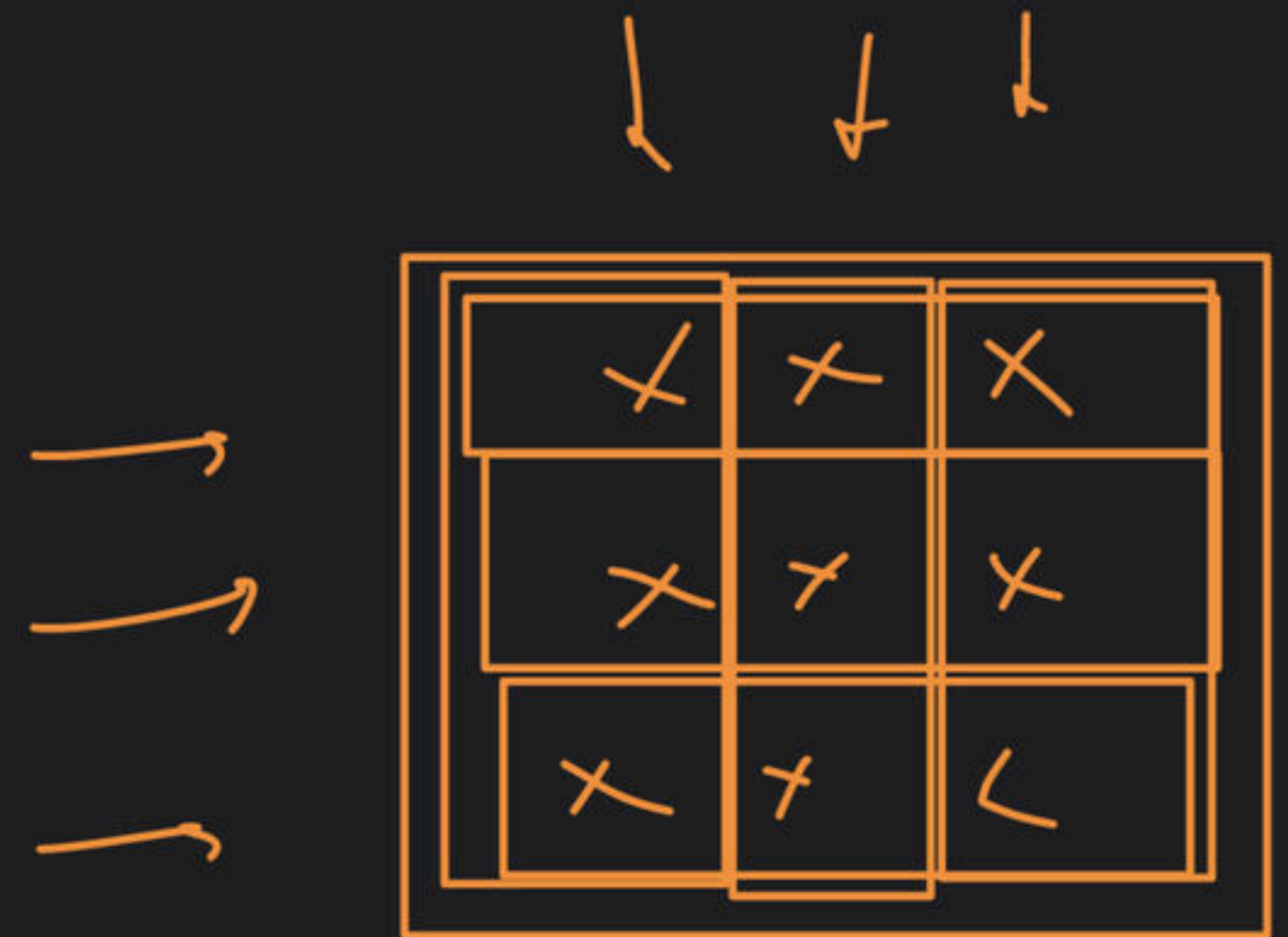
Cell

row Index $0 \rightarrow n-1$

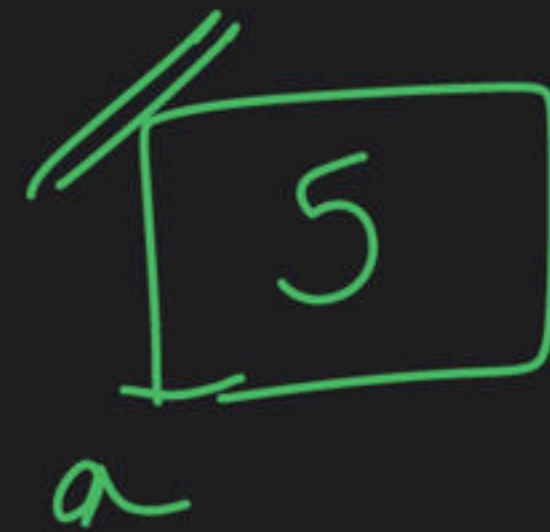
col Index $0 \rightarrow m-1$

2^d array

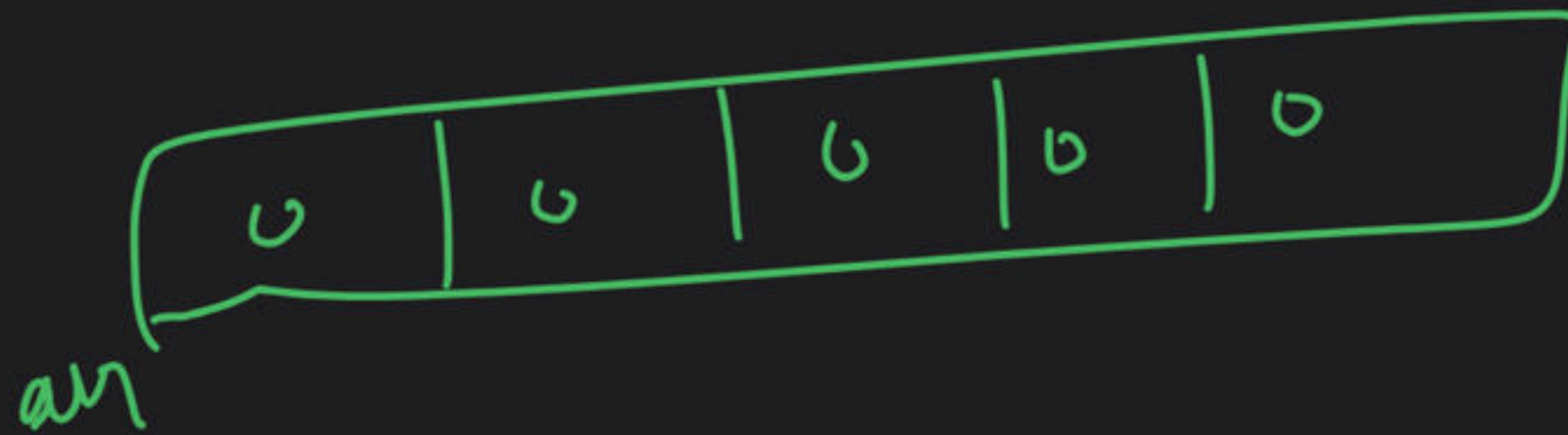
2^d vector



int a = 5

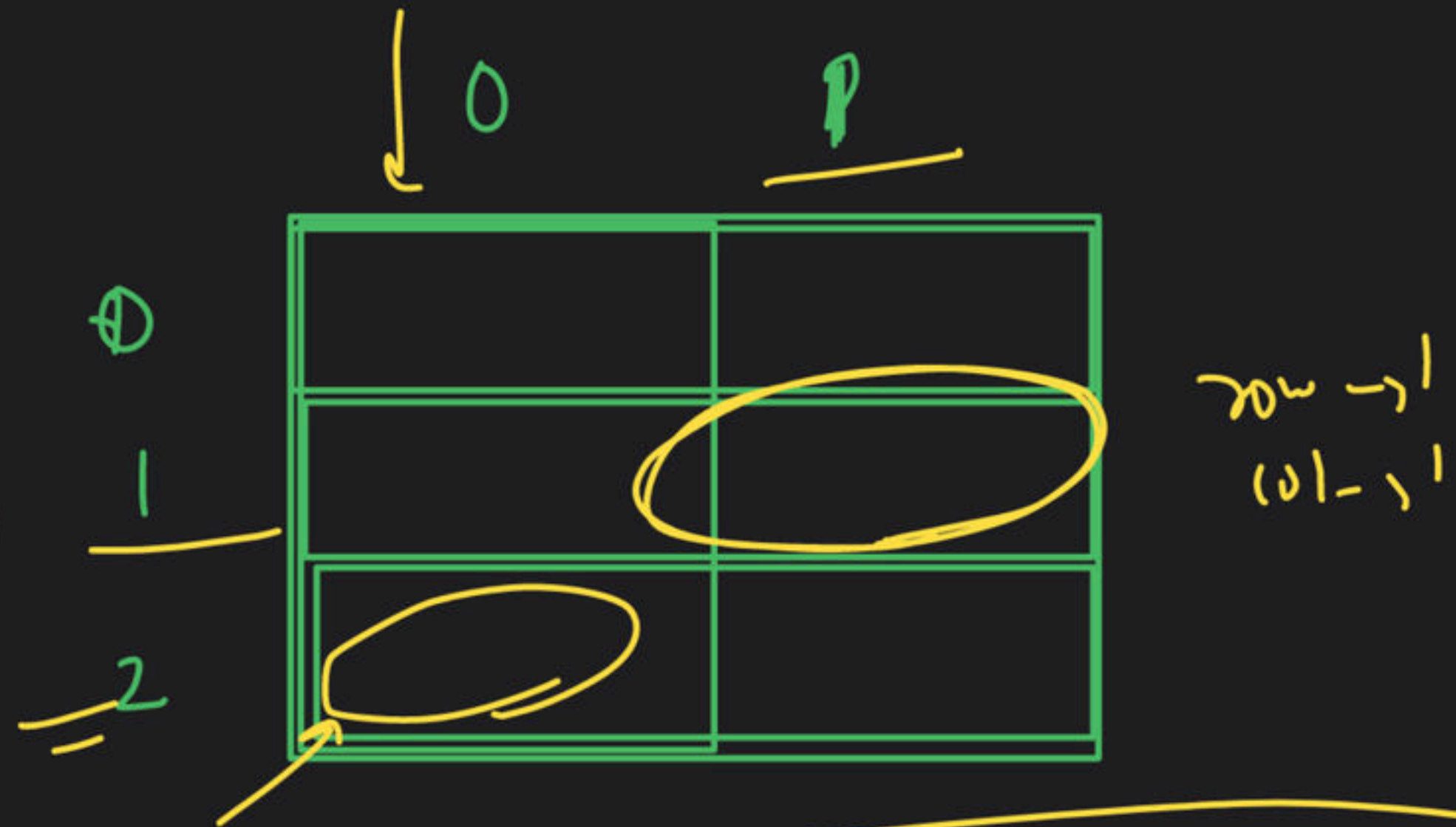


int arr[5] = {0}



`int arr[3][2];`

↓ ↓
rows columns



`arr[2][1]`

`cout << arr[1][1]`

int arr[3][3] ≤ 3

arr[2][2]

virtual
84

1	2	3
10	20	30
11	12	13

arr[2][2]

arr[0][0]

$(x \times i + j)$

$$3 \times 2 + 2 = 8$$

0	1	2	3	4	5	6	7	
1	2	3	10	20	30	11	12	13

$(x \times i + j)$

$$3 \times 0 + 12 = 12$$

~~int arr[1000][1000]~~

2 d array

create

int arr[3][4];

access

input

print

traversal

Vector

cin >> arr[i]
cin >> arr[i][j]
cout << arr[i][j]

= { { 0, 1, 2, 3, 4 },
{ 1, 2, 3, 4 },
{ 2, 3, 4, 1 },
{ 2, 3, 3, 1 } }

arr[0][0]

int rows
int cols

int [560] [117]
↙ ↘

~~arr [rows] [cols]~~

2d - array

~~1021~~

wrong

✓ c++ stl
↓
Container

vector

2d array $\rightarrow$ int arr [10] [20]
↓ ↓
row col

↓
vuv

2d

VLCADSV

→

$\text{vector} \langle \text{vector} \langle \text{int} \rangle \rangle$ arr

row

10

vector<int>(20)

haz row k andar kitni 20 column hoga

your

1st week → Orientation Session
Next

Class

Friday 9 pm

M
W
F

9 pm

Sat → 9 pm

Sun → 2 pm

$\downarrow$
 $\text{vector} < \text{vector} < \text{int} > > \text{arr} \left(\underline{\underline{10}}, \underbrace{\text{vector} < \text{int} > (20, 0)}_{\substack{\text{hr row me} \\ 20 \text{ column}}}, \uparrow_{\substack{\text{hr double} \\ \text{me } 0}} \right)$

task

-1	-1	-1
-1	-1	-1
-1	-1	-1
-1	-1	-1

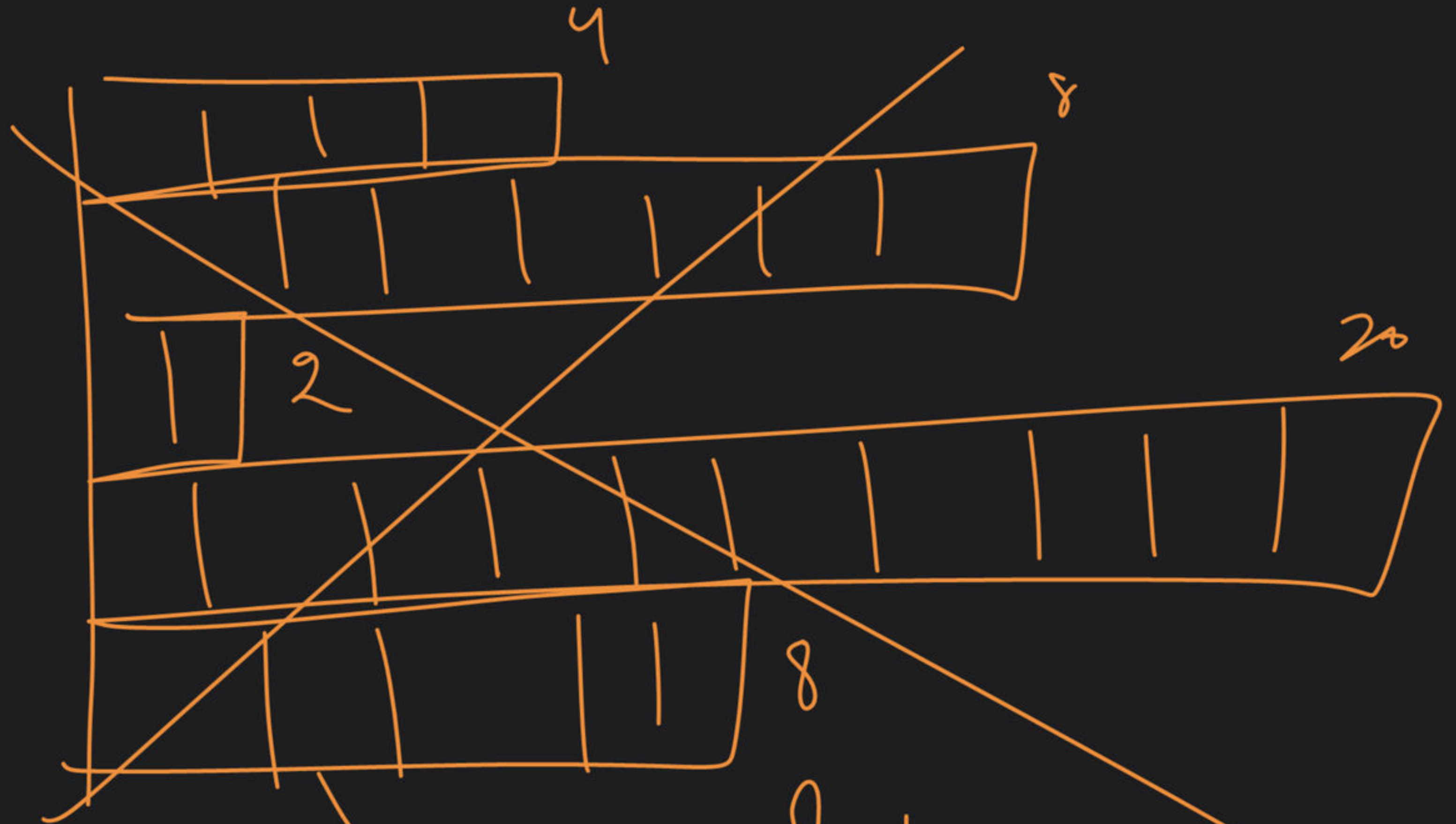
$\text{vector} < \text{vector} < \text{int} > > \text{arr} (5, \text{vector} < \text{int} > (3, -1))$

~~Staggered~~

or

~~Tagged~~

array



How → ?

Loop

H/w

1 - array

int arr[10]

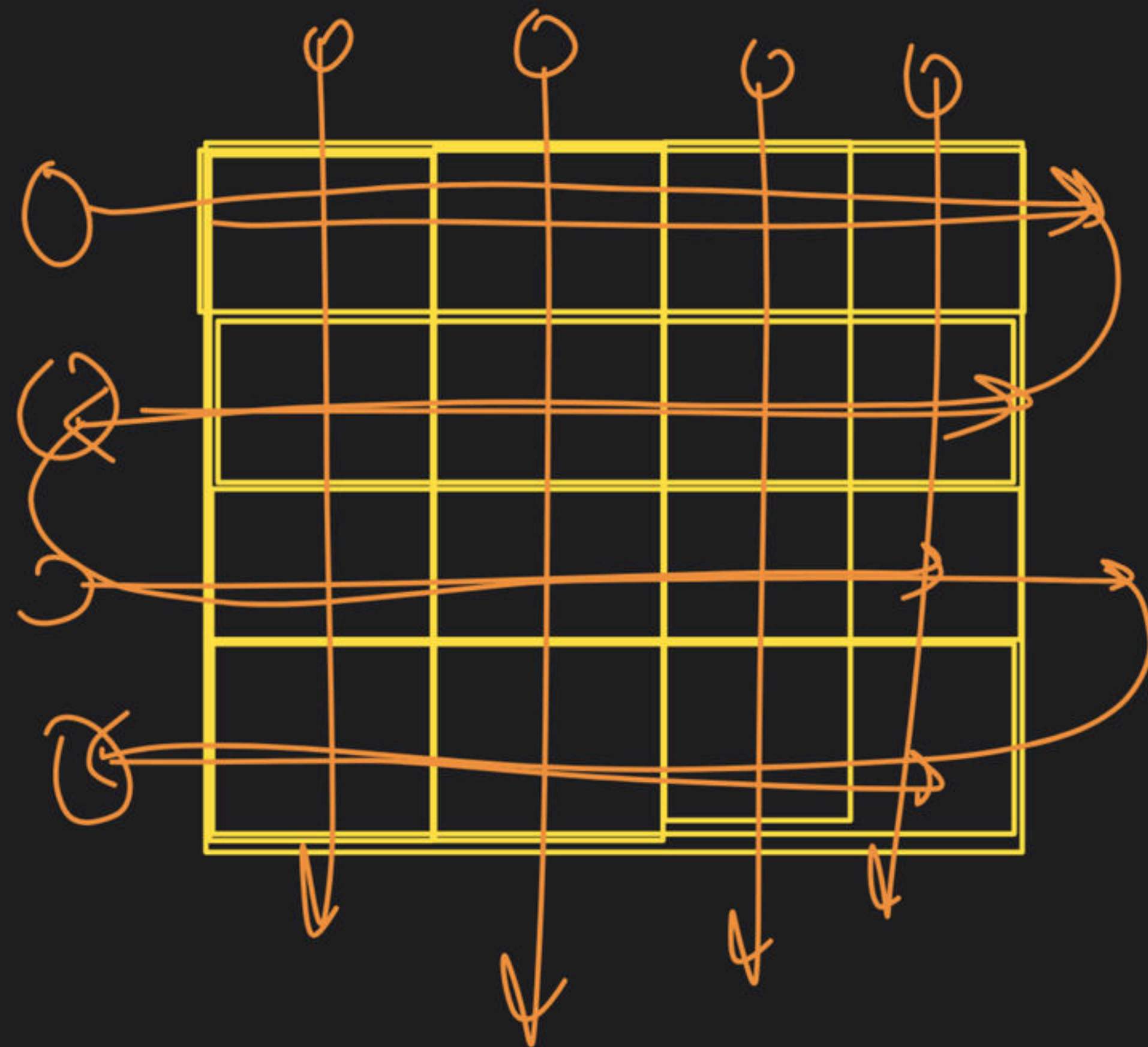
int arr[10] = {0}

1 - vector

vector<int> arr(10)

vector<int> arr(10, 0)

row-wise traversal
col-wise traversal
Snake / wave / Zig-zag
traverse
diagonal traverse
spiral traverse
Boundary traversal
→ H/W



col wis traversal

```
for ( col = 0 → < m )
```

```
{  
  for ( row = 0 → < n )
```

```
{  
    cout << arr [ col ] [ row ] ;  
}
```

```
}
```

```

for (col → 0 → < 2)
{
    for (row → 0 → < 3)
    {
        cout << arr (row) [col]
    }
}

```

$(0) = 0$
 $row = 0$
 $row = 1$
 $row = 2$



arr $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$
 row col
 0 1
 1 0
 2 0

Print
Row Wise Sum

2 min

row: 0 $\rightarrow$ Sum = 0

10	11	12
5	6	7
9	8	2

33 $\rightarrow$ print

row $\rightarrow$ Sum = 0

18 $\rightarrow$ print

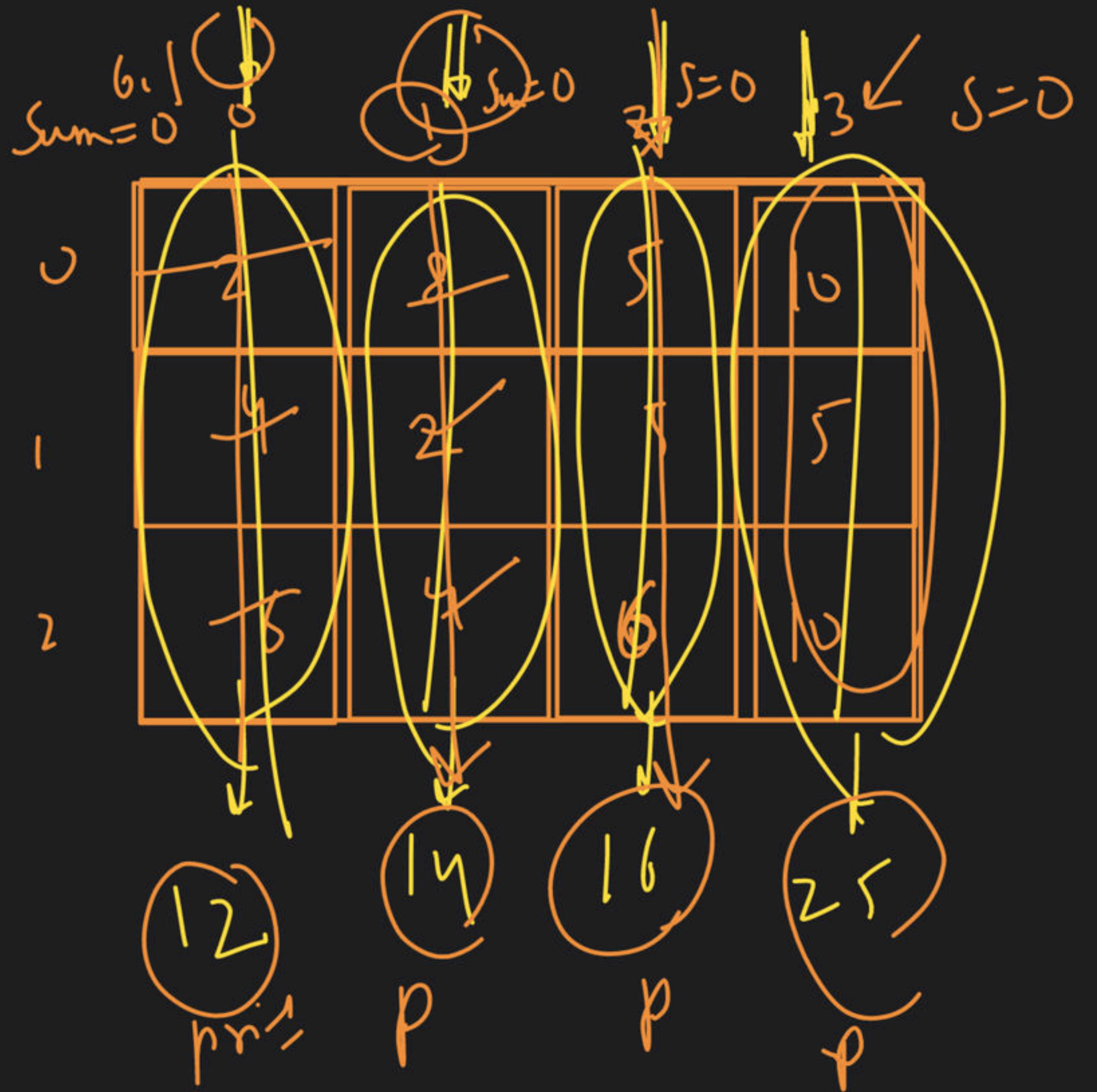
row $\rightarrow$ Sum = 0

19 $\rightarrow$ print

Print
column wise
Sum

9 min

outer
loop $\rightarrow$ col
inner $\rightarrow$ row



Search in a 2d array

2 min

i/p

target = 55

arr

1	2	10
11	21	22
33	44	55

~~find~~
~~22~~

found

return
pair (rI, cI)

pair
<-1, -1>

return

not found



10, 40, 70, 111
20, 50, 80, 12
30, 60, 90, 13

↓	↓	↓
10	20	30
40	50	60
70	80	90
11	12	13

row-wise

for ($i = 0 \longrightarrow < n$)

{ for ($j = 0 \longrightarrow < m$)

{

cout << arr[i][j]

}

}